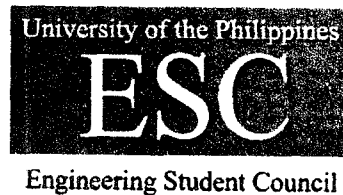


upcat reviewer

Engineering Student Council 2006-2007



ACKNOWLEDGEMENTS

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- the Engineering Student Council 2005-2006 academics committee headed by **Ms. Cherrie May Avena**
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- and the rest of the incumbent Engineering Student council officers.

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.....
NESTEE - EXY - PAUL - HAROLD - LEW - MICX - NEIL - SOL -
TANYA
.....



WHAT TO DO BEFORE THE EXAMS...

1. Know the coverage of the entrance exams in each of the schools where you are applying for admission. The usual coverage is English-Math-Science. Some schools require non-verbal tests:
2. Go over your textbooks and class notes, and attend a good review class. A review class will give you a comprehensive summary of your major high school subjects. It will also familiarize you with the different types of test questions and teach you the corresponding techniques and strategies for answering them. Thus you will feel more confident and less nervous during the actual exams. You won't waste precious time analyzing and understanding instructions. Remember, most entrance exams are under time pressure.
3. Your admission to college depends on both your high school grades (usually 40%) and your score in the entrance exams (usually 60%). You can greatly improve your chances for admission by performing well in the exams.
4. When you get your Test Permit, check if it was properly accomplished. Does it have the necessary signatures? Do you resemble your photo? Prepare your school ID - you may need it for verification of your identity.
5. Know the exact location of your Testing Center. Visit it several days before your exams so that you won't waste time looking for it right on the day of reckoning.

Check out your assigned room. Is it air-conditioned? Then bring a jacket because it's hard to take an exam when you're shivering.

Is there a CR nearby? Familiarize yourself with the location so that you will not waste time looking for it at the time of need.

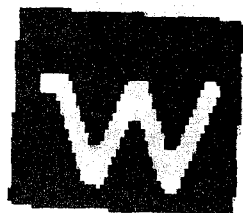
6. Prepare your pencils and erasers ahead of time. Is your pencil the prescribed brand and number? Does the eraser erase easily or does it leave smudges?
7. You will not be allowed to go out and eat during the exams.

8. Prepare yourself psychologically by being positive and optimistic. Do not allow any emotional interference before your exams. Ex., this is not the time to be quarrelling with your best friend or mother or MU!
9. Keep reviewing your textbooks, class notes, review materials and simulated tests several days before your exams.
10. Sleep well on the eve of your exams. Lack of sleep greatly impairs memory and concentration.



WHAT TO DO ON THE DAY OF YOUR EXAMS...

1. Eat a moderate meal before leaving the house. Don't eat too little - your stomach will be grumbling. Don't eat too much - your blood will rush to your digestive system, thus depriving your brain. Drink moderately so that you won't keep going to the CR.
2. Don't forget to bring your Test Permit, pencils, eraser, sharpener, and a reliable watch. The watch is very helpful in pacing yourself during the exam.
3. Leave your house early and give allowance for traffic jams. The roads and streets surrounding UP and Ateneo have massive traffic jams during entrance exams.



HAT TO DO DURING THE EXAMS...

1. Check your test booklet for any missing or misprinted pages. Ask help from the proctor or teacher if necessary
2. Listen to instructions carefully
3. Pace yourself. Don't waste time on difficult questions - skip them and go back later. But when skipping, make sure that your succeeding answers still match the questions. [i.e. make sure that your Answer no.5 is for Question no. 5]
4. Most exams are "right minus a fraction of the wrong." [e.g. a wrong answer can be given a weight of $\frac{1}{2}$ point only instead of 1 point. If you wish to guess, guess intelligently.]
5. Review your work if you still have time.
6. Be honest. Don't glance at your seatmate's answers because he may have a different set of questions.

TESTWISENESS

10 Techniques to Get a Higher Score

1. Relax. Don't be nervous. Before your exams, do this simple exercise to ease your tension: Close your eyes for a few minutes and relax your face muscles. When your face muscles relaxed, you will feel that the rest of your body will also become relaxed.
2. Know the time allotment for each segment of your exam so that you can pace yourself accordingly.

Example:

Nos. 1-60	--Science	--60 questions	--50 minutes
Nos. 61-120	--Math	--60 questions	--70 minutes
Nos. 121-180	--Reading Comprehension	--60 questions	--45 minutes
Nos. 181-250	--Language Proficiency	--70 questions	--30 minutes

3. Listen to instructions carefully. Answer the easy questions first, and then tackle the hard ones. Don't spend too much time on the hard questions - go back later if there's time.
4. Check frequently (e.g. every 10 items) to make sure that your answer matches the corresponding questions, more so after you have skipped some items. Example: Does Answer No.10 correspond to Question No. 10?
5. Look for shortcuts and avoid the length computations in solving Math problems.
6. Some questions can be answered through common sense and mental calculation.

Example:

The 140 students in the PE class were made to line up according to their class numbers, and then told to count off "1-2" and divide themselves into two lines. All the 1's were in the first line; all the 2's were in the second line. In which line was Student No. 122?

Guide: Note that all the 1's or odd-numbered students were in the first line; and all the 2's or even-numbered students were in the second line. So it's easy to conclude that Student No. 122 was in the second line.

7. Some questions are easier to solve when visualized with the use of a diagram.

Example:

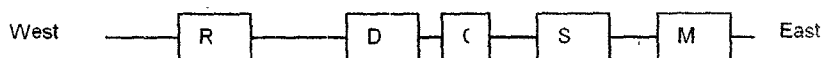
The town of Marabut is located along Marikina Highway. The town of Caranglan is west of Marabut. Sarangani is east of Caranglan, but west of Marabut. Daraga is east of Rajamon, but west of Sarangani and Caranglan. Which town is the farthest west?

First step: Draw a line to locate the relative position of each town.

"The town of Caranglan is west of Marabut." So mark their respective locations on the line.



Next Step: Place the other towns in their correct order.



Answer: The town farthest west is Rajamon.

8. Use the "Process of Elimination" (POE) when you are unsure of the right answer but sure of the wrong answers. Deciding between two choices is easier than deciding among four or five. Remember, educated guesses can raise your score, and every wrong answer you can cross out will improve your chances of guessing correctly. See your English Reviewer for examples of the use of "POE."
9. Watch out for tricks and traps. These are purposely included by the test-makers to confuse you and to test how clever you really are.

Example 1: Violation of the rule of "parallel structures"

"Eat, drink and singing -that's what men usually do in beerhouses."

Explanation: "Singing" should be "sing."

Example 2: Insertion of a clause or phrase to confuse you about subject-verb agreement

"Many politicians, in their desire to win an election, makes grandiose promises to voters."

Explanation: The subject is "politicians," so the verb should be "make."

Example 3: Use of an inverted sentence structure - the verb comes before the subject

"In the middle of vast cornfields stand the house where I was born."

Explanation: The subject is "house," so the verb should be "stands."

Example 4: Use of misplaced modifiers

"Wrapped in several layers of paper and bound securely with tape, the worker made sure that the fragile figurines would not be broken."

Explanation: This sentence gives the impression that it's the worker that was wrapped in paper and bound with tape.

10. When answering questions that determine your ability to organize phrases or sentences, look for keywords that will help your answer the question rightaway without thinking exhaustively.

Example:

- a. Secondly, they had no leader.
- b. Firstly, the workers were not united.
- c. Finally, they did not have the support of their families.
- d. Thirdly, they had no funds to tide them over.
- e. From the beginning, it looked like the workers' strike was doomed to fail.

- Answer choices:
- 1. abdec
 - 2. cbead
 - 3. deabc
 - 4. ebadc

There are several ways of approaching this question.

One way is the "Process of Elimination: "

Eliminate Answer Choice No.1 right away because it starts with "a" ("Secondly").

Eliminate Answer Choice No.2 right away because it starts with "c" ("Finally").

Eliminate Answer Choice No.3 right away because it starts with "d" ("Thirdly").

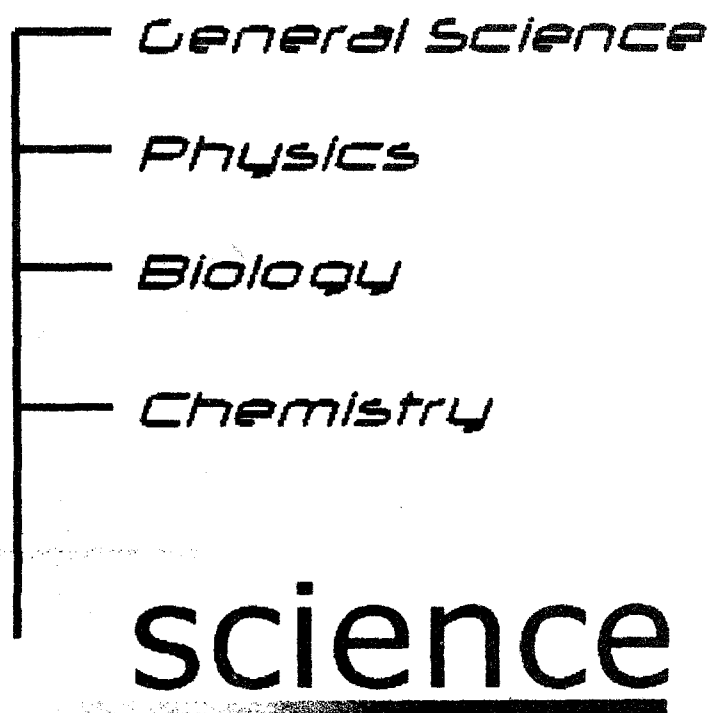
Answer Choice No. 4 is the only one left, so it must be the right choice.

Another way is by analyzing the sequence of some keywords and matching them with the answer choices:

Note the following:

"Firstly..."	is b;
"Secondly..."	is a;
"Thirdly..."	is d.

So find out which among the answer choices has the sequence "b-a-d" and you will note that only Answer Choice No. 4 contains that sequence.



GENERAL SCIENCE

Instructions: Choose the letter of the best answer from the given choices.

1. Which among these is not a good conductor of electricity?
 - a) bronze
 - b) iron
 - c) gold
 - d) plastic
 - e) none of the above
2. If the temperature of an ideal gas is increased and the volume is held constant, the pressure will
 - a) increase
 - b) decrease
 - c) remain the same
 - d) increase or decrease
 - e) cannot be determined
3. The units of carriers for codes for related proteins are called
 - a) chromosomes
 - b) genes
 - c) alleles
 - d) dominant traits
 - e) none of the above
4. The diffusion of heat through a liquid or gas by motion of its parts is
 - a) radiation
 - b) conduction
 - c) convection
 - d) greenhouse effect
 - e) none of the above
5. The shape of the earth is
 - a) oblong
 - b) sphere
 - c) oblate spheroid
 - d) cubic spheroid
 - e) none of the above

6. If both the force and area of application of the said force are halved, the pressure will
- a) increase by a factor of 2
 - b) decrease by a factor of 2
 - c) diminish by square root of 2
 - d) remain constant
 - e) none of the above
7. At what firing angle will you reach a maximum range?
- a) 30°
 - b) 60°
 - c) 45°
 - d) 90°
 - e) 10°
8. In Newton's Second Law, the ratio of the effective force applied to a particle and its acceleration is
- a) negligible
 - b) constant
 - c) unity
 - d) varying
 - e) undefined
9. Newton's Second Law was originally stated in the form
- a) $F = v dm/dt$
 - b) $F = m dv/dt$
 - c) $F = d(mv)/dt$
 - d) all of the above
 - e) none of the above
10. For a particle undergoing uniform motion, the acceleration is
- a) always nonzero
 - b) may be nonzero
 - c) always zero
 - d) always positive
 - e) cannot be determined
11. When mechanical energy is conserved, potential energy is maximum when kinetic energy is
- a) maximum
 - b) minimum
 - c) equal
 - d) cannot be determined
 - e) none of the above

12. If an object is moving at constant acceleration in a straight course, its instantaneous velocity through any time interval is
- a) greater than v_{ave}
 - b) less than v_{ave}
 - c) equal to its v_{ave}
 - d) half of its v_{ave}
 - e) none of the above
13. Given two vectors $A = 4\mathbf{m}_i + 3\mathbf{m}_j$ and $B = 2\mathbf{m}_i - 3\mathbf{m}_j$, what is $A+B$?
- a) 8.61m
 - b) 6mi
 - c) 6m
 - d) 8.61mi
 - e) none of the above
14. The atmospheric layer nearest the earth's surface is _____.
- a) Lithosphere
 - b) Stratosphere
 - c) Hydrosphere
 - d) Troposphere
 - e) none of the above
15. The most abundant of the gases composing the air is _____.
- a) Oxygen
 - b) Neon
 - c) Carbon Dioxide
 - d) Nitrogen
 - e) none of the above
16. It is the movement of molecules from a weaker liquid on one side of the barrier to a stronger liquid on the other side.
- a) Brownian movement
 - b) Osmosis
 - c) Capillary action
 - d) Diffusion
 - e) none of the above
17. According to the _____ Theory, the Earth is made up of plates moving on top of the earth's mantle. When these plates bump into each other, mountains or trenches are formed.
- a) Continental Drift
 - b) Big Bang
 - c) Volcanic Eruption
 - d) Plate Tectonics
 - e) none of the above

18. Which of the following is a non-conventional source of energy?
- a) crude oil
 - b) coal
 - c) solar energy
 - d) nuclear energy
 - e) none of the above
19. The amount of matter present in a body is called _____.
- a) volume
 - b) mass
 - c) density
 - d) weight
 - e) specific volume
20. A relationship between two living things by which one is benefited while the other is neither harmed nor benefited is called _____.
- a) Commensalism
 - b) Parasitism
 - c) Mutualism
 - d) Competition
 - e) none of the above
21. Lunar eclipse occurs when _____ comes between _____.
- a) the Moon; the Sun and the Earth
 - b) the Sun; the Moon and the Earth
 - c) Mercury; the Earth and the Moon
 - d) the Earth; the Moon the Sun and
 - e) all of the above
22. Oil can be separated from water by _____.
- a) evaporation
 - b) filtration
 - c) decantation
 - d) distillation
 - e) oil cannot be separated from water
23. The angle of the two plane mirrors that will give the most number of images is _____.
- a) less than 60 degrees
 - b) less than 45 degrees
 - c) less than 90 degrees
 - d) less than 30 degrees
 - e) less than 10 degrees

24. Which of the following activities may disturb a balanced ecosystem inside the aquarium?

- a) replacing the water
- b) feeding the fishes
- c) placing the aquarium in the porch
- d) putting too many fishes
- e) none of the above

25. Wood floats in water because

- a) its density is lesser than that of water
- b) it is porous and light
- c) its molecules are not compact
- d) its mass is lesser than that of water
- e) all of the above

26. It is a threadlike structure in the cell's molecules, which controls everything the cell does.

- a) chromosome
- b) DNA
- c) vacuole
- d) chloroplast
- e) none of the above

27. When the magma from Mayon Volcano cools and hardens, it becomes _____ rock.

- a) Igneous
- b) Volcanic
- c) Sedimentary
- d) Metamorphic
- e) none of the above

28. It is a massive ensemble of hundreds of millions of stars, all gravitationally interacting and orbiting about a common center.

- a) constellation
- b) galaxy
- c) solar system
- d) nebula
- e) supernova

PHYSICS

Instructions: Choose the letter of the best answer from the given choices.

1. The planet's motion around the sun illustrates _____.
 - a. uniform motion
 - b. accelerated motion
 - c. uniform rectilinear motion
 - d. uniform circular motion
 - e. uniformly accelerated motion
2. The period of the pendulum is affected by:
 - a. mass of the bob
 - b. length of the pendulum
 - c. total energy
 - d. time of swinging
 - e. all of the above
3. Which forces listed below cannot make two bodies attract each other?
 - a. gravitational
 - b. molecular
 - c. centrifugal
 - d. magnetic
 - e. none of the above
4. It is the change in the direction of a wave when it passes obliquely from another.
 - a. diffraction
 - b. interference
 - c. refraction
 - d. reflection
 - e. polarization
5. How much work is done in lifting a block of wood weighing 30 N to a height of 4m.
 - a. 1200 J
 - b. 120 J
 - c. 100 J
 - d. 12 J
 - e. none of the above
6. An object starting from rest accelerates at a rate of 5 m/s^2 for 10 seconds. The velocity at the end of this time is _____.
 - a. 2
 - b. 5
 - c. 50
 - d. 10
 - e. cannot be determined

7. The number of waves passing a given point each second is the wave's ____.
- a. amplitude
 - b. velocity
 - c. magnitude
 - d. frequency
 - e. pitch
8. When a ray of light strikes a mirror perpendicular to its surface, the angle of reflection is
- a. 0°
 - b. 90°
 - c. 45°
 - d. 120°
 - e. none of the above
9. Which characteristic of a wave changes as it travels across a boundary between two different media?
- a. displacement
 - b. speed
 - c. distance
 - d. power
 - e. all of the above
10. An artificial satellite making a circular orbit around the earth does not move closer to the center of the Earth because
- a. the satellite is weightless
 - b. the satellite always moves normal to the force due to gravity
 - c. the Earth's gravity keeps the satellite moving with constant velocity
 - d. a centripetal force keeps the satellite to be in its orbit
 - e. all of the above
11. Which of the following may be considered a scalar quantity?
- a. displacement
 - b. acceleration
 - c. distance
 - d. power
 - e. none of the above

12. A baseball bat exerts a magnitude of force of 60N on a ball. What is the magnitude of force exerted by the ball on the bat?
- a. 60 N
 - b. 180 N
 - c. 120 N
 - d. 20 N
 - e. none of the above
13. What is the frequency of a particular wave if its period is 0.50 seconds?
- a. 5 Hz
 - b. 1 Hz
 - c. 2 Hz
 - d. 4 Hz
 - e. 3.75 Hz
14. The acceleration of an object moving in a circular path is directed _____ the center.
- a. toward
 - b. perpendicular to
 - c. away from
 - d. tangential to
 - e. 30 degrees from
15. Liquid pressure is affected by the height and the _____ of the liquid.
- a. density
 - b. mass
 - c. area
 - d. volume
 - e. all of the above
16. A mirror with a wide radius of curvature is used as a shaving mirror in a barbershop. The face of a person looks enlarged and the skin pores are clearly seen. This mirror is called a _____ mirror.
- a. concave
 - b. convex
 - c. plane
 - d. prism
 - e. double
17. Which phenomenon of light explains the formation of images by lens?
- a. reflection
 - b. dispersion
 - c. polarization
 - d. refraction
 - e. diffraction

18. The image produced by a convex mirror is
- a. virtual and erect
 - b. virtual and inverted
 - c. real and erect
 - d. real and inverted
 - e. a convex mirror does not produce an image
19. The velocity of a car changes from 70 m/s north to 50 m/s north in 5 seconds. The magnitude of the car's acceleration is _____.
- a. -20 m/s^2
 - b. 4 m/s^2
 - c. -4 m/s^2
 - d. 14 m/s^2
 - e. 20 m/s^2
20. Which of the following heat transfers is observed when ventilating a room?
- a. radiation
 - b. conduction
 - c. convection
 - d. absorption
 - e. adsorption
21. Heat will always flow from object A to object B if object B has a lower _____.
- a. specific heat
 - b. total energy
 - c. temperature
 - d. mass
 - e. internal energy
22. If the mass of the Earth were twice as it is now, the gravitational force between it and the sun would be
- a. twice as great
 - b. thrice as great
 - c. the same
 - d. four times as great
 - e. none of the above

BIOLOGY

Instructions: Choose the letter of the best answer from the given choices.

1. What will happen to energy if food is broken down during respiration?
 - a) Absorbed
 - b) Released
 - c) Recycled
 - d) No reaction
 - e) Adsorbed
2. Which of the following could be a possible source of CO_2 ?
 - a) Decomposition
 - b) Light reaction
 - c) Evaporation
 - d) Condensation
 - e) none of the above
3. _____ is the study of heredity and variation.
 - a) Botany
 - b) Genetics
 - c) Photosynthesis
 - d) Geneticology
 - e) Zoology
4. Two populations may or may not affect each other. What do you call the interaction where in neither populations affect each other?
 - a) Mutualism
 - b) Competition
 - c) Commensalisms
 - d) Neutralism
 - e) Parasitism
5. Seasons occur because of the
 - a) shape of the earth
 - b) Sun
 - c) tilt of the earth's axis
 - d) Moon's movement
 - e) all of the above

6. Which of the following earthquake waves is the fastest and first to reach seismic station?
- a) P-waves
 - b) L
 - c) S
 - d) both S-waves and L-waves
 - e) all of the above
7. During respiration, glucose is being broken down. Which of the following is needed as raw material?
- a) Oxygen
 - b) Carbon dioxide
 - c) Water
 - d) Light
 - e) No raw material is needed.
8. The primary difference between RNA and DNA is that
- a) DNA is single-stranded and RNA is double-stranded
 - b) DNA is a double-stranded helix and RNA is single-stranded
 - c) The sugar in DNA is ribose
 - d) The sugar is both RNA and DNA is deoxyribose
 - e) DNA is not present in plants while RNA is
9. Two bacteria exchange genetic information. This process is called _____.
- a) transduction
 - b) euglena
 - c) transformation
 - d) conjugation
 - e) conjunction
10. Among the five kingdoms, protists have
- a) the fewer varieties
 - b) the least diversity
 - c) the most features in common
 - d) all of the above
 - e) none of the above

11. What differentiates a prokaryote from a eukaryote?

- a) a eukaryotic cell possesses a genetic material while a prokaryotic cell does not
- b) both have a membrane system responsible for delineating the cell from its surroundings
- c) a eukaryotic cell consists of a true nucleus that is membrane-bound while a prokaryotic cell does not
- d) algae and yeasts are examples of organisms made up of eukaryotic cells while bacteria is prokaryotic in nature
- e) none of the above

12. There are two primary types of cell division. _____ refers to the division where two daughter cells are produced from a parent somatic or body cell. Meanwhile, _____ refers to the division where four daughter cells are produced from a parent sex cell.

- a) mitosis; cytokinesis
- b) cytokinesis; mitosis
- c) mitosis; meiosis
- d) meiosis; mitosis
- e) meiosis; cytokinesis

13. In an asexual reproduction, only a single parent is involved. Which of the following is not a form of an asexual reproduction?

- a) gametogenesis
- b) binary fission
- c) fragmentation
- d) tissue culture propagation
- e) none of the above

14. This kind of interaction, also called symbiotic relationship, involves two species directly benefiting from the interaction

- a) commensalism
- b) mutualism
- c) neutral
- d) community
- e) parasitism

15. During mitosis, the correct order of the stages involved is:
- a) prephase, postphase, prophase, anaphase
 - b) prophase, telephase, metaphase, anaphase
 - c) metaphase, prophase, anaphase, telephase
 - d) anaphase, metaphase, prophase, telephase
 - e) none of the above
16. The two types of nucleic acids, deoxyribonucleic and ribonucleic, are composed of building-block materials called _____.
- a) nucleotides
 - b) organelles
 - c) cytoplasm
 - d) protein
 - e) DNA
17. Ipil-ipil seeds are used as animal feed because it has _____.
- a) fats
 - b) carbohydrates
 - c) protein
 - d) vitamins
 - e) all of the above
18. The presence of _____ distinguishes the cell as being eukaryotic
- a) cytoplasm
 - b) mitochondria
 - c) vacuole
 - d) nucleus
 - e) false feet
19. Water and other liquids enter and eventually become distributed within the plant through the _____.
- a) xylem
 - b) phloem
 - c) cambium
 - d) stomata
 - e) roots
20. Which of the following is an artificial ecosystem?
- a) herbal garden
 - b) forest
 - c) terrarium
 - d) fish pond
 - e) all of the above

21. The stage of embryonic development, which immediately follows meiotic division, is called _____.
- a) blastulation
 - b) fertilization
 - c) gastrulation
 - d) neurulation
 - e) binary fission
22. The basic unit of living matter is the _____.
- a) cell
 - b) atom
 - c) nucleus
 - d) molecule
 - e) quarks
23. Mammals are different from other animal forms because they
- a) are vertebrates
 - b) have mammary glands
 - c) have a unique skeletal system
 - d) consists of both land and water dwellers (i.e. human and whales)
 - e) have fur
24. Which of the following are found in chromosomes, which determine hereditary characteristics?
- a) cell
 - b) alkaline
 - c) ova
 - d) genes
 - e) ATP
25. What do you call the pores on the surface of a leaf?
- a) mitochondria
 - b) calyx
 - c) stomata
 - d) chromosomes
 - e) vacuoles
26. It is the process of partially sterilizing milk at high temperatures, before keeping it for some time.
- a) sterilization
 - b) evaporation
 - c) naturalization
 - d) pasteurization
 - e) skimming

27. The units of transmission of hereditary characteristics in an organism are _____.

- a) genes
- b) chromosomes
- c) off-springs
- d) dominant traits
- e) alleles

28. Why is a virus not a living organism?

- a) It can never reproduce.
- b) It is simpler than bacteria.
- c) It is not a complete cell.
- d) all of the above
- e) none of the above

29. The vascular plants include most important plants that serve as food for human beings. They are also most biologically complex of all the plants. The post-fertilization, seed-bearing remnant of a flower is a _____.

- a) fruit
- b) seed
- c) root
- d) flower
- e) none of the above

30. An advantage of sexual reproduction over asexual reproduction is that it:

- a) increases mutation rate
- b) reduces the offspring's risk of death during development
- c) increases genetic variability in a population
- d) all of the above
- e) none of the above

CHEMISTRY

Instructions: Choose the letter of the best answer from the given choices.

1. Which of the following is NOT an example of a chemical change?
 - a) souring of milk
 - b) sublimation of dry ice
 - c) rusting of iron
 - d) hair being bleached
 - e) none of the above
2. This term refers to a scientific/ intelligent guess or probable explanation to an observed phenomena or problem
 - a) conclusion
 - b) theory
 - c) Law
 - d) Hypothesis
 - e) none of the above
3. Oil and water are immiscible such that when both are placed in a single container oil "floats" on top of the water. Why is this so?
 - a) Oil is a nonpolar substance while water is polar
 - b) Oil is a polar substance while water is a nonpolar substance
 - c) Oil is less dense than water
 - d) Oil is denser than water
 - e) none of the above
4. Which of the following is NOT an example of a homogenous mixture?
 - a) air
 - b) gasoline
 - c) brass
 - d) copper
 - e) none of the above

5. _____ is an example of an _____ property

I. temperature

III. extensive

II. volume

IV. intensive

a) I, III

b) I, IV

c) IV, I

d) II, IV

e) III, I

6. The freezing point of water based on the Kelvin scale is ____ Kelvin.

a) 0°

d) 273.15°

b) 0

e) none of the above

c) 273.15

7. Being able to hit the bull's eye on a dartboard several times is an example of:

I. good precision

II. bad precision

III. good accuracy

IV. bad accuracy

a) I, III

d) II, IV

b) I, IV

e) none of the above

c) II, III

8. Which of the following statements is TRUE?

a) All atoms of an element have the same number of electrons in the nucleus of an element

b) All atoms of an element have the same number of protons and neutrons in the nucleus

c) All atoms of an element have the same number of protons in the nucleus

d) All atoms of an element have the same number of electrons and protons in the nucleus

e) none of the above

For nos. 9-12

Element X has 9 protons and 12 neutrons

9. Its atomic number is _____ which is equal to the _____

- a) 9, number of protons
- b) 12, number of neutrons
- c) 9, number of electrons
- d) 21, sum of the number of protons and neutrons
- e) none of the above

10. Its mass number is _____

- a) 9
- b) 12
- c) 18
- d) 21
- e) none of the above

11. An atom that has 9 protons but 15 neutrons is a/an _____ of Element X.

- a) isomer
- b) isotope
- c) example
- d) particle
- e) none of the above

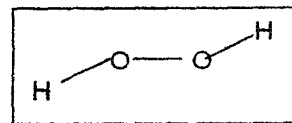
12. Which of the following does not belong to the group: Oxygen, Hydrogen, Nitrogen, Iodine, Boron, Fluorine

- a) Iodine
- b) Boron
- c) Fluorine
- d) Nitrogen
- e) Hydrogen

13. Given the structural formula of hydrogen peroxide is given shown

in the figure at the right. Its empirical formula is given by _____

while its molecular formula is _____



- a) $2\text{HO}; \text{OH}$
- b) $2\text{H}_2\text{O}_2; 2\text{HO}$
- c) $\text{H}_2\text{O}_2; \text{OH}$
- d) $\text{OH}; \text{H}_2\text{O}_2$
- e) $\text{OH}; 2\text{HO}$

14. Na^+ is an example of a/an _____ while Cl^- is an example of a/an _____
- a) cation; anion
 - b) cation; reactant
 - c) anion; nonmetal
 - d) anion; metal
 - e) none of the above
15. Metal atoms tend to _____ while nonmetal atoms tend to _____
- a) lose electrons; gain protons
 - b) gain protons; lose electrons
 - c) lose electrons; gain electrons
 - d) gain electrons; lose protons
 - e) none of the above
16. This Law states that a given compound has a constant relative numbers and kinds of atoms
- a) Law of Conservation of Mass
 - b) Law of definite/ constant composition
 - c) Law of constant compound
 - d) Law of atomic structure
 - e) none of the above
17. Which of the following chemical equations does not belong to the group?
- a) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
 - b) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
 - c) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 - d) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
 - e) none of the above
18. Given the unbalanced equation: $\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ What type of reaction is given above, which is a rapid reaction that produces flame?
- a) single displacement
 - b) decomposition
 - c) combination
 - d) combustion
 - e) double displacement

19. As determined experimentally by Italian scientist Amadeo Avogadro, one mole of any substance represents _____ particles

- a) 2.06×10^{23}
- b) 6.02×10^{26}
- c) 6.02×10^{23}
- d) 2.60×10^{23}
- e) 6.20×10^{23}

20. Acetone, which is used as solvent and as a reactant in the manufacture of Plexiglass, boils at 56.1°C . What is the temperature in Fahrenheit scale?

- a) 101°F
- b) 159°F
- c) 114°F
- d) 133°F
- e) 63.2°F

21. Aside from the properties of the molecules involved, the principal factors that affect the speed of a chemical reaction are:

- I. presence of catalyst
 - II. the temperature at which the reaction occurs
 - III. the concentration of the reactants
 - IV. turbulence in the reacting medium
- a) II, III
 - b) I, III
 - c) III
 - d) III, IV
 - e) none of the above

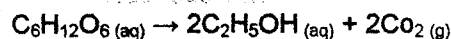
22. Which does not belong to the group?

- I. Making observations
 - II. Testing a hypothesis
 - III. Drawing conclusions
 - IV. Scientific method
- a) I
 - b) II
 - c) III
 - d) IV
 - e) none of the above

23. Solid calcium carbide, CaC_2 , reacts with water to form an aqueous solution of calcium hydroxide and acetylene gas C_2H_2 . The balanced chemical equation corresponds to:

- a) $\text{CaC}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{CaOH} + \text{C}_2\text{H}_2(\text{g})$
- b) $2\text{CaC}_2(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Ca}(\text{OH})_2 + 2\text{C}_2\text{H}_2(\text{g})$
- c) $3\text{CaC}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 3\text{Ca}^{2+} + 3\text{OH}^- + 3\text{C}_2\text{H}_2(\text{g})$
- d) $\text{CaC}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{aq}) + \text{C}_2\text{H}_2(\text{g})$

24. The fermentation of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, produces ethyl alcohol, $\text{C}_2\text{H}_5\text{OH}$ and CO_2 :

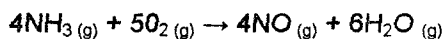


How many grams of glucose are needed to produce 92g of ethyl alcohol?

- a) 2
- b) 1
- c) 360
- d) 180
- e) none of the above

For nos. 25-28,

In an experiment, 5.00g of ammonia reacts with 3.000g of oxygen



25. Which is the limiting reactant? Excess reactant?

- a) O_2 ; NO
- b) NH_3 ; H_2O
- c) O_2 ; NH_3
- d) NH_3 ; O_2
- e) none of the above

26. How many grams of NO form?

- a) 2.25
- b) 2.2
- c) 2
- d) 8.78
- e) none of the above

33. Consider the elements of group 2A, which of the following is correct according to increasing atomic size?

- a) Hydrogen, Lithium, Sodium
- b) Magnesium, Calcium, Strontium
- c) Calcium, Barium, Strontium
- d) Sodium, Lithium, Potassium
- e) Calcium, Magnesium, Strontium

34. What is the correct electron configuration of Chloride ion? $\text{Cl} = 17 \rightarrow \text{Cl}^- = 18$

The atomic number of chlorine is 17.

- a) $1s^2 2s^2 2p^6 3s^2 3p^5$
- b) $1s^2 2s^2 2p^6 3s^2 3p^4$
- c) $1s^2 2s^2 2p^6 3s^2 3p^6$
- d) $1s^2 2s^2 2p^6 3s^1 3p^5$
- e) $1s^2 2s^2 2p^6 3s^2 3p^5 4s^1$

35. What kind of substance turns blue litmus paper to red?

- a) acid
- b) base
- c) salt
- d) metal
- e) metalloid

36. If the pressure exerted on a confined ideal gas is doubled, what happens to its temperature and volume, taken separately?

- a) doubled; doubled
- b) remains the same
- c) halved; halved
- d) halved; doubled
- e) double; halved

37. Which of the following statements is/are true?

- I. The tendency of the universe is to go towards minimum entropy or disorder
- II. Energy can neither be created nor destroyed
- III. The reaction is said to be spontaneous when $\Delta G > 0$
- IV. The internal energy of a system is a state function

a) I and II

c) II and IV

b) I and III

d) II, III, and IV

e) II only

38. In the reaction $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$, _____ is reduced while _____ is oxidized.

a) Zn^{2+} , Cu

d) Cu, Zn^{2+}

b) Zn, Cu^{2+}

e) None of the above

c) Cu^{2+} , Zn

39. In a certain experiment, different volumes (listed below) of 1M hydrochloric acid (I) and 1M sodium hydroxide (II) are mixed.

Solution	I (ml)	II (ml)
Solution A	75	25
Solution B	50	50
Solution C	25	75

What can be said about Solution B?

a) It is warm and toxic

b) Solution B is less acidic than Solution A.

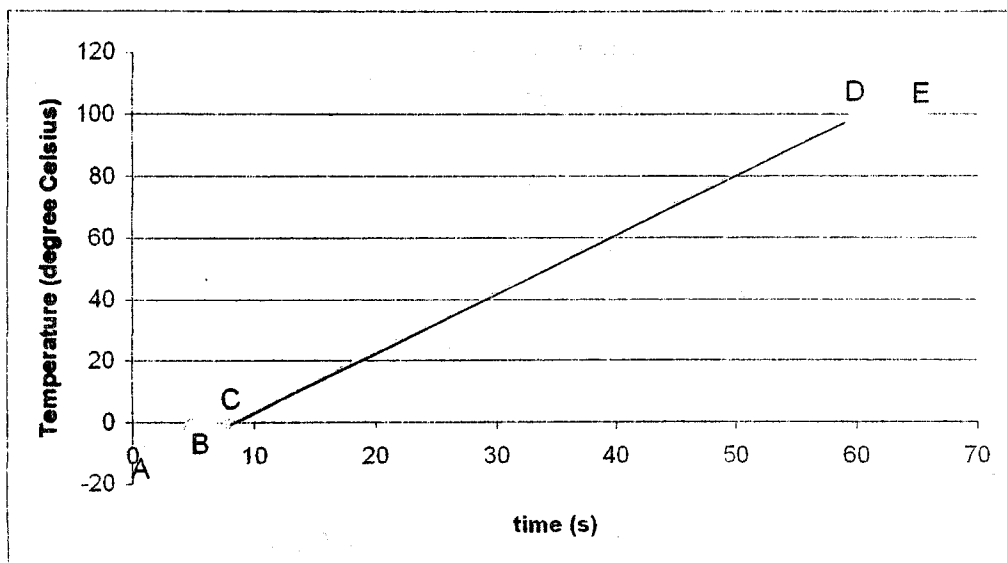
c) The solution has a pH of greater than seven.

d) All of the above

e) None of the above

For nos. 40-41

Given the temperature profile of water below at 1 atm.



40. If the above graph is based from experiment, which of the following observations is/are true?

- I. water is being heated from points A to E
- II. The entropy change at points B to C and D to E is greater than the entropy change at points A to B and C to D.
- III. Line BC corresponds to the normal freezing point of water
- IV. Where the temperature remains constant there is no energy added or released

- | | |
|-------------------|----------------------|
| a) I, II | d) I, II, III |
| b) I, II, IV | e) None of the above |
| c) I, II, III, IV | |

41. What will happen to line DE if the pressure is changed from 1 atm to 10 atm?

- a) The slope of line DE will be negative
- b) The slope of line DE will be positive
- c) The slope of line DE will remain constant at the same temperature of 100°C
- d) The slope of line DE will remain constant but at a temperature greater than 100°C
- e) The slope of line DE will remain constant but at a temperature less than 100°C

42. The Ideal Gas Law is applicable only to a certain extent because Real gas molecules have finite volumes and attract one another. What are the conditions wherein deviations from ideal gas behavior can be observed?
- a) High pressure
 - b) Low temperature
 - c) High temperature
 - d) Low pressure
 - e) None of the above

- Simple Calculus*
- Geometry*
- Trigonometry*
- Statistics*
- Word Problems*

math

SIMPLE CALCULUS and ALGEBRA

Instructions: Choose the letter of the best answer from the given choices.

1. Given $h(x) = \sqrt{4-x^2}$. What is the domain of h ?
 - a) $\mathbb{R}$
 - b) $[-2, 2]$
 - c) $(2, -2)$
 - d) none of the above
2. What is the range of h ?
 - a) $\mathbb{R}$
 - b) $[-2, 0]$
 - c) $[0, 2]$
 - d) $(-2, 2)$
3. $f(x) = |x|$. What is the range of x ?
 - a) $\mathbb{R}$
 - b) $[-\infty, 0]$
 - c) $[0, +\infty]$
 - d) none of the above
4. $\lim_{x \rightarrow 5} (3x-7) = ?$
 - a) 8
 - b) 7
 - c) 6
 - d) 5
5. Suppose $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{1, 4, 9, 16\}$. Then $A \cap B = ?$
 - a) $\{1, 2, 4, 6, 8, 9, 10, 12, 16\}$
 - b) $\emptyset$
 - c) $\{4\}$
 - d) $\{4, 16\}$

For nos. 6-10

Determine if the following statements are (a) TRUE, or (b) FALSE.

6. if $a < b$ and $b < c$, then $a < c$
7. if $a < b$ and $c < 0$, then $ac < bc$
8. if $a < b$ and $c < d$, then $a + c < b + d$
9. $\sqrt{(5^2)} = |5|$
10. $|a/b| = |a|/|b|$ for all $a, b \in \mathbb{R}$

20. $\lim_{x \rightarrow 1^-} \sqrt{1-x^2} = ?$

a) 0.5

b) -0.5

c) 0

d) does not exist

21. Given $f(x) = \begin{cases} 2, & x \leq 3 \\ -2, & x > 3 \end{cases}$ What is the $\lim_{x \rightarrow 3^+} f(x)$?

a) 2

b) -2

c) 0

d) does not exist

22. What is the $\lim_{x \rightarrow 3^-} f(x)$?

a) 2

b) -2

c) 0

d) does not exist

23. What is the $\lim_{x \rightarrow 3} f(x)$?

a) 2

b) -2

c) 0

d) does not exist

24. $|2x - 3| \leq 6$. Solve for x.

a) $\{-3/2, 9/2\}$

b) $(-3/2, 9/2)$

c) $[-3/2, 9/2]$

d) $[3/2, 9/2]$

25. $f(x) = 3/\sqrt{x^2 - 1}$. What is the domain of f?

a) $1 > x > -1$

b) $x > 1$ and $x < -1$

c) $1 \geq x \geq -1$

d) $x \geq 1$ and $x \leq -1$

26. Divide $12x^3y^4 + 18x^4y^2 - 36xy^3$ by $3x^2y^2$

a) $4xy^2 + 6x^2 - 12y$

b) $4xy^2 + 6x^2 - 12y/x$

c) $4xy^2 + 6x^2 + 12y$

d) $4xy^2 + 6x^2 + 12y/x$

27. Multiply $2x - 3y$ by $3x + 4y$

a) $6x^2 + xy - 12y^2$

b) $6x^2 - 17xy - 12y^2$

c) $6x^2 - xy - 12y^2$

d) $6x^2 + 17xy - 12y^2$

28. $(3x + 2y)(9x - 6xy + 4y) = ?$

a) $27x^3 + 8y^3$

c) $27x^3 + 6xy + 8y^3$

b) $27x^3 - 6xy + 8y^3$

d) $27x^3 - 30xy + 8y^3$

29. Factor $9x^2 - 30xy + 25y^2$

a) $(3x - 5y)(3x + 5y)$

c) $(3x + 5y)^2$

b) $(3x - 5y)^2$

d) $(5x - 3y)(5x + 3y)$

30. Reduce $(8x^4y^7) / (12x^6y^3)$ to lowest terms

a) $2/3$

c) $(2x^2y) / (3xy)$

b) $(2x^2) / (3y^4)$

d) $(2y^4) / (3x^2)$

31. Simplify: $a^2/b - b^2/a$

a) $a - b$

c) $a^2b - ab^2$

b) $(a^3 - b^3) / ab$

d) $a^3 - b^3$

32. Simplify: $(7a / 12b^3) * (20b^5 / 35a^3)$

a) $a^2 / 3b^2$

c) $b^2 / 3a^2$

b) $3a^2 / b^2$

d) $3b^2 / a^2$

33. Simplify: $(a^{-1} + b^{-1}) / (a + b)^{-1}$

a) $1 / (a + b)^2$

c) $1/a + 1/b$

b) $a + b$

d) $(a + b)^2 / ab$

34. $81^{-3/4} = ?$

a) 27

c) $\sqrt{27}$

b) $1/27$

d) $1/\sqrt{27}$

35. $\sqrt[6]{27} = ?$

a) 3

c) $\sqrt{3}$

b) $1/3$

d) $1/\sqrt{3}$

36. $4\sqrt{3} - 5\sqrt{12} + 2\sqrt{75} = ?$

a) $4\sqrt{3}$

c) $\sqrt{3}$

b) $-6\sqrt{3} + 6\sqrt{5}$

d) $\sqrt{5}$

44. For all $x \neq \pm 2$, $(2x^2 + 6x + 8) / (x^2 - 4)$ is equal to:

- a) $(2x + 3) / (x - 4)$
- b) $(2x^2 + 6x + 8) / (x - 2)^2$
- c) $2(x^2 + 3x + 3) / [(x - 2)(x + 2)]$
- d) $(x^2 + 3x + 3) / [(x + 1)(x - 4)]$

For nos. 45-50

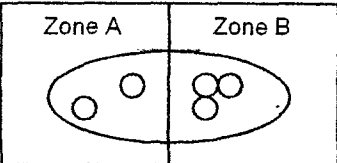
Which of the following is greater: column A or column B?

A: Column A is greater

B: Column B is greater

C: A and B are equal

D: cannot be determined

	Column A	Column B
45. $f(x) = 3x + 1$ $g(x) = 2x$	$(f \circ g)(x)$	$(g \circ f)(x)$
46. $x = -97$	$\sqrt{x^2}$	$ x $
47. $-1 < n < 0$	$1/n$	n^3
48. Five students take test Q	median of their scores	mean of their scores
49. N = set of natural numbers	number of elements of N	no. of even nos. in N
50. 	$(\# \text{ of fish in A}) / (\# \text{ of fish in B})$	$1/\pi$

For nos. 51-55

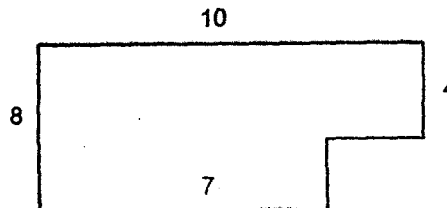
Compare the quantities in column A and in column B and determine if

- a) $A > B$
- b) $B > A$
- c) $A = B$
- d) No comparison

Column A	Column B
51. three successive discounts of 5% each	single discount of 15%
52. $3x+23+100$	$3y+100000$
53. area of a square with perimeter 8	area of a circle having radius 2
54. the percent decrease from 12 to 10	the percent decrease from 22 to 20
55. $2/x^2$	$2/x^5$

6. Find the area of the following figure. All measurements are in cm

- a) 68 cm^2
- b) 56 cm^2
- c) 10 cm^2
- d) Cannot be determined



7. How many foci does an ellipse have?

- a) 2
- b) 1
- c) 4
- d) 0

8. A straight line whose endpoints are any two points of a circle

- a) Chord
- b) Radius
- c) Circumference
- d) Center

9. Planar coordinate system where a point is described by its distance from two perpendicular straight lines.

- a) Planes
- b) 3D objects
- c) Cartesian
- d) Polar

10. What is the distance of these two endpoints, $(3, -2)$ and $(-3, -2)$?

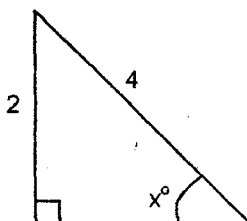
- a) 3.5
- b) 6
- c) 0
- d) None of the above

TRIGONOMETRY

Instructions: Choose the letter of the best answer from the given choices.

1. From the following figure, find x°

- a) 60°
- b) 90°
- c) 30°
- d) 10°



2. If $\tan \Theta = -1/2$ and $270^\circ \leq \Theta \leq 360^\circ$, what is $\sin \Theta$?

- a) $-1/\sqrt{5}$
- b) $1/\sqrt{5}$
- c) $1/2$
- d) cannot be determined

3. Find the quadrant containing angle x° wherein $\sin x^\circ < 0$ and $\tan x^\circ > 0$

- a) 1st quadrant
- b) 2nd quadrant
- c) 3rd quadrant
- d) 4th quadrant

4. If $\cos \Theta = -1/2$ and $\cot \Theta > 0$, find $\tan \Theta$

- a) $1/2$
- b) $-\sqrt{3}/2$
- c) $\sqrt{3}$
- d) $-1/2$

5. Find the value of $\sin 17\pi/4$

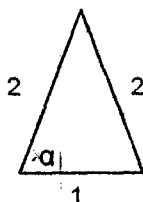
- a) $1/2$
- b) $\sqrt{3}/2$
- c) $-\sqrt{2}/2$
- d) $1/\sqrt{2}$

6. Express $\tan^2 x \csc x$ in terms of $\sin x$

- a) $\sin^2 x / (1 - \sin^2 x)$
- b) $\sin x / (\sin^2 x - 1)$
- c) $\sin x / (1 - \sin^2 x)$
- d) cannot be determined

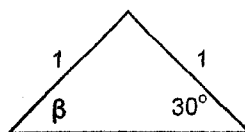
7. Given the figure on the right, find $\cos \alpha$.

- a) $\frac{1}{2}$
- b) $\frac{1}{4}$
- c) $\frac{1}{3}$
- d) 0



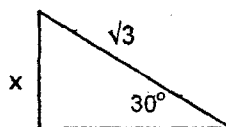
8. Given the figure on the right, find angle β .

- a) 90°
- b) 20°
- c) 60°
- d) 30°



9. Given the figure on the right, find x .

- a) $\frac{1}{2}$
- b) $-\frac{1}{2}$
- c) $\frac{\sqrt{3}}{2}$
- d) 0



10. Which of the following is a false statement?

- a) $\tan \theta = \sin \theta / \cos \theta$
- b) $\cot \theta = \sin \theta / \cot \theta$
- c) $\sin^2 \theta + \cos^2 \theta = 1$
- d) $\csc^2 \theta - 1 = \cot^2 \theta$

11. Express 30° into radians.

- a) 0.54
- b) 0.52
- c) 1
- d) cannot be determined

12. $\cot \theta / \tan \theta =$ _____

- a) 1
- b) $\sin \theta / \cos \theta$
- c) $\cos^2 \theta / \sin^2 \theta$
- d) $\sin^2 \theta / \cos^2 \theta$

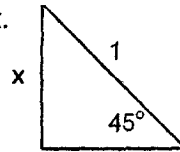
13. Given the figure on the right, find x .

a) $\sqrt{3}/2$

b) $\frac{1}{2}$

c) $\sqrt{2}/2$

d) $-1/2$



STATISTICS

Instructions: Choose the letter of the best answer from the given choices.

1. A managerial accountant's sample of six companies' book value shows \$25, \$7, \$22, \$33, \$18, \$15. The sample mean is:
 - a) 19
 - b) 20
 - c) 18
 - d) 21
2. If $\bar{x}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2 , then the limit form of the distribution of $Z = (\bar{x} - \mu) / \sigma / \sqrt{n}$ as n approaches infinity, is the standard normal distribution $N(0,1)$. What do you call this theorem?
 - a) Probability theorem
 - b) Bell-shaped
 - c) Poisson
 - d) Central Limit Theorem
3. The set of all possible outcomes / events of a statistical experiment is called
 - a) Element
 - b) Sample space
 - c) Tree diagram
 - d) Observation
4. A _____ is an arrangement of all or part of a set of objects
 - a) Combination
 - b) Probability
 - c) Mutually exclusive
 - d) Permutation
5. How many possible combinations are there if I have 3 shorts, 4 blouses and 5 pair of shoes?
 - a) 60
 - b) 50
 - c) 12
 - d) none of the above
6. Find the mean and median of the numbers from 1 to 10.
 - a) 5.5, 5.5
 - b) 5.5, 6
 - c) 5.5, 5
 - d) 10, 5.5

14. What is the probability that at least one tail occurs when a coin is tossed twice?

a) $\frac{3}{4}$

c) 2

b) 1

d) $\frac{1}{4}$

15. From a group of 4 men and 5 women, how many committees of size 2 are possible if there are no restrictions?

a) 126

c) 36

b) 24

d) none of the above

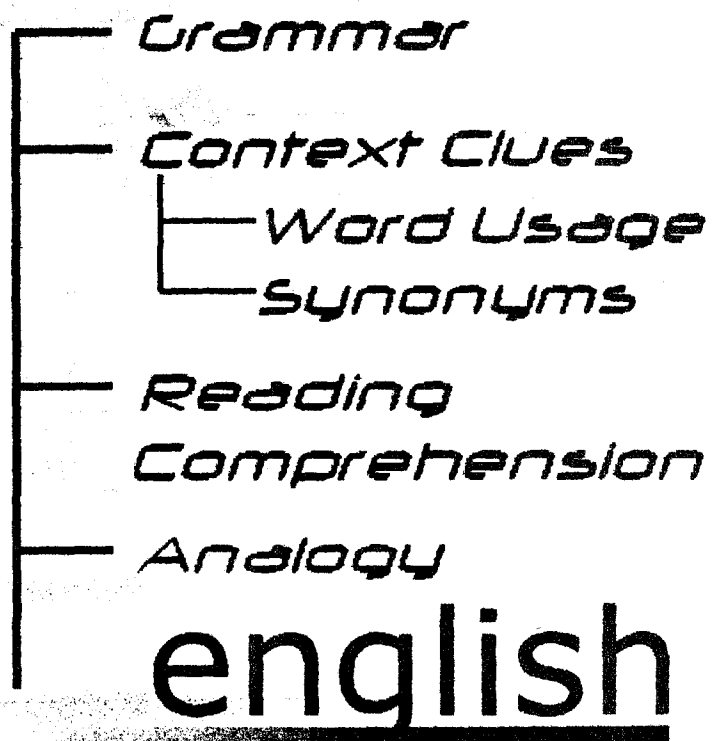
WORD PROBLEMS

Instructions: Choose the letter of the best answer from the given choices.

1. What is the sum of the first 20 counting numbers, the first 20 even counting numbers and the first 20 odd numbers?
a) 2300
b) 650
c) 10001
d) 790
e) none of the above
2. If May deposited her money in the bank who has an annual interest of 50% compounded annually last year and has plans to buy a \$6000 bike, when should she withdraw her money if he deposited \$4000?
a) 2 years from now
b) 10 years from now
c) next year
d) this year
e) never
3. If Ju can do a painting in 7 hrs, May can do the job in 8 hrs. How many hrs will they do 9 painting jobs if they do it at the same time?
a) 33 hrs 17 mins
b) 33 hrs 40 mins
c) 33 hrs 36 mins
d) none of the above
e) cannot be determined
4. A car is traveling due north. At the same time the car left, a truck followed but with slower speed. If the car travels 60 km/hr and the truck travels 23 km/hr, at what time will the car be 100 km away from the truck?
a) 2 hrs, 42 min, 9.7 sec
b) 1 hr, 18 min, 7.5 sec
c) 3 hrs, 23 min, 11.4 sec
d) 7 hrs, 11 min, 2.1 sec
e) none of the above

10. What's the velocity of the water coming out of the hose if the nozzle is quarter circle in cross section with 3-inch diameter?
- a) $0.64\pi^2$ ft/s
b) $0.64\pi^2$ in/s
c) $0.32\pi^2$ ft/s
d) $0.32\pi^2$ in/s
e) none of the above
11. Jason can build a table in 12 hrs. Trisha can also build the same table in 9 hrs. Joe can build the same table in 11hrs. In a day, Jason starts making tables at 9 am. Trisha joined him at 11 am and Joe came at 12 noon. How many days will they consume if they are tasked to make 30 tables with a daily schedule show above while ending at 5 pm everyday?
- a) 33 days
b) 25 days
c) 49 days
d) 17 days
e) none of the above
12. Ben is going to play lotto tomorrow. He only plays 1 combination every time he plays a bet. What is his chance in winning the jackpot if the game chooses 4 numbers out of the 20 and must be in a particular order for the jackpot to be won?
- a) 116280
b) 1/116280
c) 1/25240
d) 25240
e) none of the above
13. A plane travels due north at 8 am with a speed of 100 km/hr. another plane travels 60 degrees east of the first plane at the same time with 120 km/hr speed. How far will they be from each other after 6 hrs?
- a) 3.03 km
b) 10.20 km
c) 668 km
d) 220 km
e) none of the above

18. Noong nakaraang limang taon, ang edad ni Ana ay pitong beses ng edad ng kanyang anak. Sampung taon mula ngayon, ang edad niya ay doble ng edad ng kanyang anak. Ano ang kanilang edad sa kasalukuyan?
- a) Ana = 26; anak = 8
 - b) Ana = 40; anak = 14
 - c) Ana = 51; anak = 18
 - d) Ana = 24; anak = 7
 - e) Ana = 32; anak = 10
19. May and Che work together in preparation for a project. They can finish a job in 6 hours if they work together. After 3 hours of working together, May has to leave Che because she has an unexpected date. It takes Che 9 hours to finish the remaining job alone. How long will it take each of them to finish a job alone?
- a) Che = 18 hours; May = 9 hours
 - b) Che = 20 hours; May = 10 hours
 - c) Che = 9 hours; May = 18 hours
 - d) Che = 10 hours; May = 20 hours
 - e) Che = 15 hours; May = 15 hours
20. Si Ryan ay nag-invest ng P 1,000,000, isang bahagi ay sa may interest na 9% at ang natitira ay sa 7%. Ang kabuuang halaga na kanyang interest ay P 86,000. Magkano ang ininvest niya sa may 9% na interest?
- a) 100 000
 - b) 800 000
 - c) 600 000
 - d) 400 000
 - e) 200 000



GRAMMAR

Instructions: Select the letter that corresponds to the underlined word/s that make the sentence grammatically incorrect. Select the letter 'E' if there are no grammatical errors found.

1. Everyone in our family except my brother and I has eaten lunch. No error.

A B C D E

2. Hearing the news the whole group went to the province. No error.

A B C D E

3. The chairman of the board consulted his members about the committee's major problem.

A B C D

No error.

E

4. None of the players in the locker room were able to comprehend or to solved the problem.

A B C D

No error.

E

5. Science refer to that dynamic, cumulative systems of verifiable concepts and processes

A B C

which seek to understand natural phenomena. No error.

D E

6. From the given choices, select the word that appropriately complete each set of relationship.

A B C D

No error.

E

7. Preprocessor directives can appear anywhere in a source file, but they apply only to the

A B C D

remainder of the source file. No error.

E

8. The cover photograph is a computer rendered image of the building along Makati. No error.

A B C D E

However, their interest in improving the teaching of automotive courses have not subsided.

A B C D

No error.

E

10. Sign has been used, not only to enhance the quality of illustrations but also to help students

A

B

distinguish the difference between the two. No error.

C

D

E

11. A number of optional sections have been included. No error.

A

B

C

D

E

12. As he travels on the curved portion of a track, each racer are subjected to an acceleration

A

B

C

directed toward the center. No error.

D

E

13. The professor came to her and said, "Laura, how often do you study our subject." No error.

A

B

C

D

E

14. To enjoy all rewards and privileges, please present these card when purchasing at the store.

A

B

C

D

No error.

E

15. She ignore the comment and settled in at her terminal. No error.

A

B

C D

E

16. Everyone except Ernie and Bert have seen the film. No error.

A

B

C

D

E

17. Finally he winked up at Karel, and then both his eyes closed. No error.

A

B

C

D

E

18. The lieutenant smiles and seemed to enjoy the Commissioner's sense of irony. No error.

A

B

C

D

E

19. He felt like he was flying though; and that was good enough. No error.

A

B

C

D

E

20. Sitting along the beach across the clinic, Lito wondered what he was suppose to do. No error.

A

B

C

D

E

21. The suspect demanded for a 20 minute break before the trial resumes. No error.

A

B

C

D

E

22. At the intersection of Raft Avenue and Stock Street, the road dead-ends at a vast two storey

A

B

C

D

shed. No error.

E

23. Both of his bodyguards move forward, but the minister waved them off. No error.

A

B

C

D

E

24. Cherry began to point east across Park Avenue when she was struck by the flat of the man's hand. No error.

A B s C D

E

25. Wilbert and Isaiah Jones was dragging badly when they left the Plaza later that night.

A B C D

No error.

E

26. I only come to ask you if you would bring your Holy Water to the graveyard. No error.

A B C D E

27. I guess he sat down resting in the shade and never got up again. No error.

A B C D E

28. What attitude towards life does the speaker seem to be expressing? No error.

A C D E

29. Her childhood was happy, and she was content enough when she became a young woman.

A B C D

No error.

E

30. The second master, in appreciation of his wit and learning gave him his freedom. No error.

A B C D E

31. That night, he cradled the half frozen calf in his great arms and carried it home. No error.

A B C D E

32. Everyone in the class understood the project except I, and that is why I feel so stupid.

A B C D

No error.

E

33. Please call me when their already in the meeting place. No error.

A B C D E

34. The oil-rich country of Brunei lays to the south of the Republic of the Philippines. No error.

A B C D E

35. When I called you up this morning, what are you doing? No error.

A B C D E

CONTEXT CLUES

WORD USAGE

Instructions: Determine which of the following synonyms on the given choices best corresponds to how the underlined word/s were used on each sentence.

1. Leo made a disappointed, misunderstood face, from the many creases at his disposal.
 - a) authority
 - b) distribution
 - c) garbage unit
 - d) regulation
2. Ultimate stress design should be used only for projects when the enforcement of the provisions of the construction specifications is well observed.
 - a) strengthening
 - b) implementation
 - c) regulation
 - d) putting of force
3. Joanna's reply to his query was sure and crisp, that it didn't took too long before she was able to prod him away.
 - a) brittle
 - b) lively
 - c) concise
 - d) fragile
4. The first permanent tooth usually appears about four years after the baby teeth have erupted.
 - a) exploded
 - b) broken out
 - c) burst out
 - d) emerged
5. The professor took an immaculate handkerchief out of his pocket, and began to polish the lenses of his glasses.
 - a) pure
 - b) spotless
 - c) error-free
 - d) flawless
6. By the end of the century, mass circulation of newspapers had been developed.
 - a) movement of blood through veins and capillaries
 - b) dissemination
 - c) distribution of copies
 - d) availability in libraries

7. Artifice was one of the qualities of style that was worthy of further cultivation.
- a) fostering of growth
 - b) refinement
 - c) development
 - d) elevation
8. Much of his writing is slack that is why readers get bored at his works.
- a) lacks action
 - b) lacks completeness/ perfection
 - c) lacks firmness
 - d) lacks diligence
9. Members shall be adequately braced to resist lateral and torsional displacement.
- a) connected together
 - b) supported with braces
 - c) reinforced
 - d) get ready for an attack
10. Biscuits had been stamped on the floor, making mopping more laborious than it already is.
- a) crushed
 - b) impressed
 - c) pounded
 - d) imprinted
11. Please depress the red button when you're ready.
- a) press down
 - b) repress
 - c) discourage
 - d) weaken
12. The commander ordered his men to hold their posts no matter what happens.
- a) support
 - b) maintain
 - c) adhere to
 - d) carry
13. When Cherrie and Chester finally got into the elevator, they found themselves crushed against each other.
- a) powdered to small pieces
 - b) silenced
 - c) squeezed
 - d) had an infatuation
14. One week later, the vessel deposited them in one of the biggest ports on the Eastern Peninsula.
- a) consigned
 - b) dropped
 - c) credit
 - d) accumulated

15. Pauline's words had made all sorts of frightened echoes in our minds. The word echo in this sentence means
- a) a close imitation
 - b) an answering sympathetic effect
 - c) a soft recurrence of a statement
 - d) a reverberation of sound waves
16. The gases trapped in pasty lava **do not escape freely** and may sometimes break out with explosive violence.
- a) sudden
 - b) expressive
 - c) unrestrained
 - d) volatile
17. The dead load consists of the weight of the structure completed.
- a) empty
 - b) inert
 - c) constant
 - d) lifeless

SYNONYMS

Instructions: Choose the letter of best answer from the given choices.

18. Terrie kept fumbling the materials, gathering a few and shuffling them as if they were cards.
- a) playing with
 - b) juggling
 - c) groping for
 - d) moving clumsily
19. Two police officers sprawled over a couch in Johanna's living room, as if they were in their own homes.
- a) creep awkwardly
 - b) lay down
 - c) jumped into
 - d) sit with arms and legs spread out

20. The interviewer, dark, young and serious, turned his attention to us with a cramped sort of face.

- a) paralyzed
- b) cheerful
- c) bothered
- d) expressionless

21. The dog ambled to the door, which was slightly ajar, and slowly squeezed out into the passage.

- a) crept
- b) ran
- c) walked
- d) jumped

22. Harold lumbered at him and clouted him across the head.

- a) cover with a clout
- b) hit forcefully
- c) stroke gently
- d) kissed

23. I thought they were likely to bundle him into a car but the policemen simply led him back to the road and pointed him towards the town center.

- a) wrap
- b) hustle
- c) a lot of things
- d) group together

24. The brass bars are bundled in two's, three's and four's.

- a) wrapped
- b) hustled
- c) a lot of things
- d) grouped together

25. The tractor-trailer backed away from the smoldering wreck and stopped.

- a) burning steadily
- b) flaming increasingly
- c) burning with much smoke
- d) exploding

26. The gymnast plunged from bar to bar, twisting and turning with dazzling speed.

- a) skipped playfully
- b) thrust forward
- c) swung lightly
- d) dangled idly

27. The Ifugao lady gently slipped off the headband that held the basket high on her back so that she could rest for a while.

- a) removed quickly
- b) slid out of place
- c) wore in
- d) threw away

READING COMPREHENSION

Instructions: Each passage below is followed by questions based on its content. Choose the best answer and shade the box corresponding to that answer.

Passage 1:

Ho-ti trembled every joint while he grasped the abominable thing, wavering whether he should not put his son to death for an unnatural young monster, when the crackling scorched his fingers, as it had done his son's, and applying the same remedy to them, he in his turn tasted some of its flavor, which, make what sour mouths he would pretence, proved not altogether displeasing to him. In conclusion (for the manuscript here is a little tedious) both father and son fairly sat down to the mess, and never left off till they had dispatched all that remained of the litter.

(An excerpt from Charles Lamb's *A Dissertation Upon Roast Pig*)

1. In the first sentence, the nearest meaning to the word "abominable" could be

a) humongous	c) illegal
b) detestable	d) dangerous
2. What remedy did the father used for his scorched fingers?

a) he put his fingers under the running water	b) he went outside and buried them in snow
c) he licked them	d) he wrapped cloth around his fingers
3. What kind of figure of speech was used in the first statement, "Ho-ti trembled every joint"?

a) simile	c) hyperbole
b) metaphor	d) personification

Passage 2:

In a sense we have come to our nation's capital to a cash check. When the architects of our republic wrote the magnificent words of the Constitution and the Declaration of Independence, they were signing a promissory note to which every American was to fall heir. This note was a

promise that all men, yes, black men as well as white men, would be guaranteed the unalienable rights of life, liberty, and the pursuit of happiness. It is obvious today that America has defaulted on this promissory note, insofar as her citizens of color are concerned. Instead of honoring this sacred obligation, America has given the Negro people a bad check, a check has come back marked "insufficient funds."

(An excerpt from Martin Luther King Jr.'s *I Have A Dream*)

4. The word "defaulted" in the passage means
- | | |
|----------------------|-----------------------------|
| a) set for automatic | c) remove fault |
| b) neglected | d) failed to appear on time |
5. The whole passage is developed by
- | | |
|----------------------------|---------------------|
| a) comparison and contrast | c) cause and effect |
| b) narration | d) dialogue |
6. The purpose of the passage is to
- | | |
|--------------------------|--------------|
| a) humor | c) preach |
| b) bring about awareness | d) enlighten |

Passage 3:

Sun Tzu said: There are five ways of attacking with fire. The first is to burn soldiers in their camp; the second is to burn stores; the third is to burn baggage trains; the fourth is to burn arsenals and magazines; the fifth is to hurl dropping fire amongst the enemy.

In order to carry out an attack, we must have means available. The material for raising fire should always be kept in readiness. There is a proper season for making attacks with fire, and special days for starting a conflagration.

The proper season is when the weather is very dry; the special days are those when the moon is in the constellations of Sieve, the Wall, the Wing or the Cross-bar; for these four are all days of rising wind.

(An excerpt from Sun Tzu's *The Art of War*)

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12. When do you consider a man full?

- a) when he can offer
- b) when he can read
- c) when he can see
- d) when he can read and understand

13. How can a reader learn from a book?

- a) when he joins arguments
- b) when he knows many writers
- c) when he can write poems
- d) when he applies and assimilates the words

14. According to the passage, why does a man read?

- a) to contradict and confute
- b) to believe and take for granted
- c) to waste and swallow
- d) to weigh and consider

15. How will a reader treat a book?

- a) as a friend
- b) as a confidante
- c) as a memoir
- d) as a historian

Passage 5:

Peter came to him and said, "Lord, how often shall my brothers sin against me, and I forgive him? Up to seven times?"

Jesus said to him, "I say to you, not seven times, but seventy times seven."

Thus the kingdom of heaven is like a certain king who had servant that owed him ten thousand talents. Because he could not pay, the lord ordered him to be sold, and his wife and children, and all that he had, and payment be made. The servant therefore fell down and worshipped him saying, "Lord have patience with me, and I will pay you all." Then the

Lord of that servant was move with constant compassion and freed him and forgave him the debt.

"But the same servant went out and found one of his fellow servants who owed him a hundred pence, and he laid hands on him, and took him by the throat, saying, "Have patience with me and I will pay you all." And he would not, but went and cast him into prison, till he pays his debt.

So when his fellow servants saw what was done, they were very sorry and came and told their lord all that was done.

"Then his lord, after calling him forth said to him 'O wicked servant, I forgave you all that debt, because you begged me. Should you not also have the compassion on your fellow servant, even as I had pity on you?' And his lord was angered and delivered him to the torturers, till he should pay all that was due to him. So likewise shall my heavenly Father do also to you, if you do not forgive from your heart your brothers' trespasses."

16. Jesus told Peter that he must forgive his brother

- | | |
|------------------------|---------------|
| a) seven times | c) many times |
| b) seventy times seven | d) sometimes |

17. The king forgave his servant because

- a) he felt sorry for him
- b) the servant paid him quickly
- c) the servant fell down and hurt himself
- d) the servant's wife promised to sell their goods to pay the debt

18. When the servant was imprisoned by the man who owed him money, his companion

- a) did nothing
- b) were angry
- c) asked him what he was doing
- d) reported what he had done to their lord

19. In this parable, Jesus is teaching about

- | | |
|----------|----------------|
| a) money | c) forgiveness |
| b) kings | d) prisons |

20. The servant's lord was angry with him finally because

- a) he had not paid his debt
- b) he had no pity on his fellow servant
- c) the lord wanted money
- d) the servant continued to beg for more time to pay

ANALOGY

Instructions: Select the word or pair of words that appropriately completes the following relationship.

1. ODE : POETRY; _____ : _____
 - a) THEME - ESSAY
 - b) CONFLICT - SHORT STORY
 - c) FABLE - PROSE
 - d) TRAGEDY - DRAMA
 - e) DENOUEMENT - NOVEL
 2. EACH : SOMEBODY; _____ : _____
 - a) THAT - OH
 - b) GRACEFULLY - DANCE
 - c) SING - SONG
 - d) WHICH - WHOM
 - e) MAN - HE
 3. PEAL : PEEL; _____ : _____
 - a) STEAL - STILL
 - b) FAIR - FARE
 - c) BUY - VIE
 - d) GREAT - GREET
 - e) MEAL - KNEEL
 4. HERD : CATTLE; _____ : _____
 - a) COLONY - TERMITES
 - b) PACK - PEACOCKS
 - c) HIVE - WOLVES
 - d) SWARM - QUAIL
 - e) COVEY - WHALES
 5. EPIC : ELEGY; _____ : _____
 - a) ALLEGRO - PLOT
 - b) SONATA - SATIRE
 - c) RADAR - ALLEGORY
 - d) DISSECT - PROTAGONIST
 - e) COUPLET - SONNET
- GOLF : HOLE; _____ : _____
- a) BASEBALL - GLOVE
 - b) ARCHERY - BULL'S EYE
 - c) BOWLING - PIN
 - d) TENNIS - RACKET
 - e) BASKETBALL - CENTER

7. PROTEIN : MEAT; _____ : _____
- a) ENERGY - SUGAR
 - b) CALORIES - WATER
 - c) VITAMINS - MINERALS
 - d) STARCH - POTATOES
 - e) FAT - CHOLESTEROL
8. FERN : PLANT; _____ : _____
- a) SHARK - WATER
 - b) CAT - LION
 - c) COLLIE - BOXER
 - d) GERM - DISEASE
 - e) DOG - ANIMAL
9. TRY : TRIED; _____ : _____
- a) CAME - COME
 - b) FLY - FLEW
 - c) GO - GONE
 - d) DEAD - DIED
 - e) SIT - SITTING
10. LION : CUB; _____ : _____
- a) DUCK - DRAKE
 - b) CHILD - BABY
 - c) HORSE - COLT
 - d) GOAT - LAMB
 - e) DEER - VENISON
11. RABIES : DOGS; _____ : _____
- a) MALARIA - MOSQUITOES
 - b) FEVER - ANTS
 - c) COUGH - WORMS
 - d) AIDS - RAT
 - e) LEPROSY - BUTTERFLIES
12. GLOVES : HAND; _____ : _____
- a) WELDER - STEEL
 - b) GUN - COMBAT
 - c) FIELD - FOOTBALL
 - d) HELMET - HEAD
 - e) ARMY - SOLDIER
13. GUN : HOLSTER; _____ : _____
- a) SWORD - SCABBARD
 - b) SWORD - SLAY
 - c) SWORD - BAYONET
 - d) SWORD - CADET
 - e) SWORD - WAR

14. CANVAS : PAINTING; _____ : _____

a) CLAY - POTTERY

d) CLAY - STATUE

b) CLAY - MOLD

e) CLAY - BAKE

c) CLAY - ART

15. IRON : SOLID; _____ : _____

a) WATER - GASEOUS

b) WATER - PAIL

c) WATER - COMPOUND

d) WATER - ELEMENT

e) WATER - FLUID



ANSWER KEY

SCIENCE

General Science

1. d
2. a
3. b
4. c
5. c
6. d
7. c
8. b
9. c
10. b
11. b
12. c
13. b
14. d
15. d
16. b
17. d
18. c
19. b
20. a
21. d
22. c
23. d
24. d
25. e
26. a
27. a
28. b

Physics

1. d
2. b
3. c
4. c
5. b
6. c
7. d
8. a
9. b
10. d
11. c
12. a
13. c
14. a
15. a
16. a
17. d
18. a
19. b
20. c
21. c
22. a

Biology

1. b
2. a
3. b
4. d
5. c
6. a
7. a
8. b
9. d
10. e
11. c, while b&d are true, the question remains unanswered. (always choose the best and most appropriate answer)
12. c
13. a
14. b
15. e, prophase, metaphase, anaphase, telephase
16. a
17. c
18. d
19. b
20. c
21. a
22. a
23. b
24. d
25. c
26. d
27. a
28. a
29. a
30. d

Chemistry

1. b, sublimation : solid to liquid
2. d
3. c
4. d, air is gaseous soln , gasoline= liquid soln, brass=solid, copper=element
5. b, intensive property: does not depend on the amt of substance/ sample
6. c, $^{\circ}\text{K}$ is absolute zero, the lowest attainable temp. Kelvin does not use $^{\circ}$
7. a
8. c
9. a
10. d
11. b
12. b the rest normally exists as diatomic molecules
13. d
14. a
15. c
16. b
17. b
18. d
19. c
20. d, $^{\circ}\text{F} = 9/5^{\circ}\text{C} + 32$
21. e I,II,III
22. d, I, II, and III are parts/steps in the scientific method (which is the general approach to problems)
23. d
24. d
25. c
26. a
27. e
28. c, following rules of significant figures
29. b, $0.1 = [(x \text{ g})(\text{moles}/ 158 \text{ g})] / (200\text{L})$ temperature
30. a, $M_1V_1 = M_2V_2 \Rightarrow 12V_1 = 1 \times 0.6$
31. a
 $\text{C}_2\text{H}_6 + 7/2 \text{ O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$
 $3.5 \text{ mol ethane} \times (4 \text{ moles CO}_2 / 2 \text{ moles ethane}) = 7 \text{ moles CO}_2$
 $2.0 \text{ mol O}_2 \times (4 \text{ mole CO}_2 / 7 \text{ moles O}_2) = 1.1 \text{ moles CO}_2$
32. a, $0.5/1.1 = 0.45$
33. b
34. c
35. a
36. e
37. c
38. c, Cu^{2+} , Zn
 LEORA = lose electrons Oxidation (gets oxidized) reducing agent
 GEROA = gain electrons Reduction (gets reduced) oxidizing agent
39. Answer: Given the statements the best answer would be b. ph of Sol'n B is 7 since equal volumes of HCl and NaOH react. All three solutions undergo a neutralization reaction with water and a salt (specifically NaCl) as product.
40. d
41. d, Boiling point Elevation. Boiling point temperature increases with increasing external pressure.
42. e, nota. High pressure and low temperature

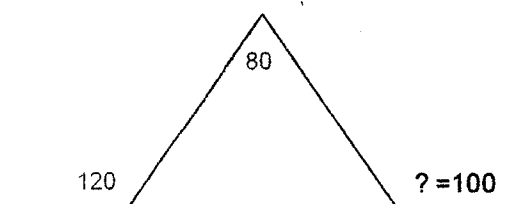
MATHEMATICS

Simple Calculus

1. b, $4 - x^2 \geq 0$
 $x^2 \leq 4$
 $x \leq 2$ and $x \geq -2$
2. c
3. d
4. a, $\lim_{x \rightarrow 5} 3x - \lim_{x \rightarrow 5} 7$
 $= 3 \cdot 5 - 7 = 8$
5. c
6. a
7. b
8. a
9. a
10. b, $b \neq 0$
11. a, $x = (5 + 1)/2$; $y = (-3 - 6)/2$
 $x = 3$; $y = -9/2$
12. d, $m = (-1 - 3)/(5 - 5) = -4 / 0$
undefined
13. b, $m = (3 - (-3)) / (-2 - 6)$
 $= -3/4$
 $y - (-3) = (-3/4)(x - 6)$
 $4y + 12 = -3x + 18$
 $3x + 4y - 6 = 0$
14. b
15. c
16. c
17. a
18. d
19. d
20. c
21. b
22. a
23. d
24. c, $-6 \leq 2x - 3 \leq 6$
 $-3 \leq 2x \leq 9$
 $-3/2 \leq x \leq 9/2$
 $x = [-3/2, 9/2]$
25. b, $\sqrt{x^2 - 1} > 0$
 $x^2 - 1 > 0$
 $x^2 > 1$
 $x > 1$ and $-1 > x$
26. b, $(12x^2y^4 / 3x^2y^2) + (18x^4y^2 / 2x^2y^2) -$
 $(36x^2y^2) = 4xy^2 + 9x^2 - 36x^2y^2$
27. c, $(2x - 3y)(3x + 4y)$
using the FOIL method
F: $2x \cdot 3x = 6x^2$
O: $2x \cdot 4y = 8xy$
I: $-3y \cdot 3x = -9xy$
L: $-3y \cdot 4y = -12y^2$
 $6x^2 - xy - 12y^2$
28. a, $(3x+2y)[(3x)^2 - (3x)(2y) + (2y)^2]$
 $= (3x)^3 + (2y)^3 = 27x^3 + 8y^3$
29. b, perfect square: $(3x-5y)^2$
30. d, $8x^4y^7 / 12x^6y^3 = 2y^4 / 3x^2$
31. b, $(a^2 \cdot a - b^2 \cdot b) / (ab) = (a^3 - b^3) / (ab)$
32. c, $(7/35)(20/12)(a/a^3)(b^5/b^3)$
 $= (1/5)(5/3)(1/a^2)(b^2)$
 $= b^2 / 3a^2$
33. d, $(1/a + 1/b) / [1/(a+b)]$
 $= (b+a)(a+b)/(ab)$
 $= (a+b)^2 / (ab)$
34. b, $(3^4)^{3/4} = 3^3 = 1/27$
35. c, $\sqrt[6]{(3^3)} = 3^{3/6} = 3^{1/2} = \sqrt{2}$
36. a, $4\sqrt{3} - 5\sqrt{4 \cdot 3} + 2\sqrt{25 \cdot 3}$
 $= 4\sqrt{3} - 10\sqrt{3} + 10\sqrt{3} = 4\sqrt{3}$
37. b, $(2^2)^{3x} = (2^4)^{x+1}$
 $2^{6x} = 2^{4x+4}$
 $\log_2 2^{6x} = \log_2 2^{4x+4}$
 $2x = 4$
 $x = 2$
38. c, $x^4 = 1/16$
 $x = \sqrt[4]{(1/16)}$
 $x = 1/2$
39. c, $(x+3)(x-3) / (x-3) = y$
 $x = y - 3$
40. c
41. b
42. b
43. c
44. c
45. b
46. c
47. b
48. d
49. c
50. a
51. b
52. d
53. b
54. a
55. d

Geometry

- a, Surface area = $4\pi(2^2) = 50.26$
- a, Solution $200\text{cm} = 2\text{m}$
 $V = (1/3)(B)(h) = (1/3)(4)(2) = 2.67 \text{ m}^3$
- a, Supplement = $180 - 37 = 143$,
Complement = $90 - 37 = 53$
- d, Since AB and CD is not parallel and no sufficient data, we cannot determine the answer.
- b,



- a, total area = $8(10) = 80 \text{ cm}^2$,
small rectangle area = $3 \times 4 = 12 \text{ cm}^2$
Area = $80 \text{ cm}^2 - 12 \text{ cm}^2$
- a, 2
- a, chord
- c
- b

Statistics

- b
- d
- b
- d
- a
- a, mean =
 $(1+2+3+4+5+6+7+8+9+10)/10$
 $= 5.5$
median = $(5+6)/2 = 5.5$
- c, (7 is the most frequent element)
- c
- d
- a
- c
- b
- c
- a
- d

Trigonometry

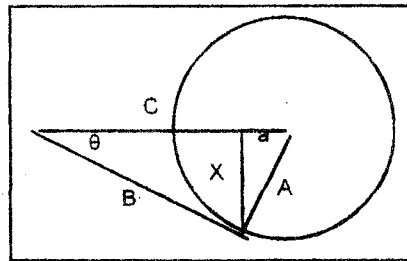
- c, $\sin x = 2/4$
 $\sin x = 1/2$
 $x = \sin^{-1} 1/2 = 30^\circ$
- a, $\sin \theta / \cos \theta = -1/2$
 $-2 \sin \theta = \cos \theta$
 $\sin^2 \theta + \cos^2 \theta = 1$
 $\sin^2 \theta + 4 \sin^2 \theta = 1$
 $5 \sin^2 \theta = 1$
 $\sin \theta = \pm 1/\sqrt{5}$
since $270^\circ \leq \theta \leq 360^\circ$
 $\sin \theta = -1/\sqrt{5}$
- c, $\sin x < 0 = 3^{\text{rd}} \text{ Q and } 4^{\text{th}} \text{ Q}$
 $\tan x = 1^{\text{st}} \text{ Q and } 3^{\text{rd}} \text{ Q}$
Therefore, 3^{rd} Q
- c, $\cos \theta = x/r$
 $1 + y^2 = 4$
 $y^2 = 3$
 $y = -\sqrt{3}$
 $\cot \theta > 0$ at the 3^{rd} Q
 $P(-1, -\sqrt{3})$
 $\tan \theta = -\sqrt{3} / -1 = \sqrt{3}$
- d, $\sin 17\pi/4 = \sin(\pi/4 + 4\pi)$
 $= \sin \pi/4 = 1/\sqrt{2}$
- c, $\tan^2 x \csc x = (1/\sin x) * (\sin^2 x / \cos^2 x)$
 $= \sin x / \cos^2 x$
 $= \sin x / (1 - \sin^2 x)$
- b, $c^2 = a^2 + b^2 - 2ab \cos \alpha$
 $4 = 4 + 1 - 2*2*1*\cos \alpha$
 $4 - 5 = -2*2*1*\cos \alpha$
 $-1 = -4*\cos \alpha$
 $\cos \alpha = 1/4$
- d, $\sin \beta / 1 = \cos \beta / 1$
 $\sin \beta = \cos \beta$
 $\beta = \sin^{-1} 1/2$
 $\beta = 30$
- c, $\sin 30 = x/\sqrt{3}$
 $1/2 = x/\sqrt{3}$
 $x = \sqrt{3}/2$
- b, $\cot \theta = \cos \theta / \sin \theta = 1 / \tan \theta$
- b, $30 * \pi / 180 = 0.52$
- c, $\cot \theta = 1 / \tan \theta$
 $(1 / \tan \theta) / \tan \theta = 1 / \tan^2 \theta$
 $= 1 / (\sin^2 \theta / \cos^2 \theta)$
 $= \cos^2 \theta / \sin^2 \theta$

Word Problems

1. e. $n(n+1)/2 + m(m+1)/2 = (20)(20+1)/2 + (40)(40+1)/2 = 1030$
2. d. $F = P(1+i)^N$
 $6000 = 4000(1+0.50)^N$
 $\log 6000 = \log [4000(1+0.50)^N]$
 $\log 6000 = \log 4000 + \log[(1+0.50)^N]$
 $\log 6000 = \log 4000 + N \log(1+0.50)$
 $N = (\log 6000 - \log 4000) / [\log(1+0.50)]$
 $N = [\log(6000/4000)] / [\log(1+0.50)]$
 $N = \log 1.5 / \log 1.5$
 $N = 1$
 This year
3. c. $(1/8 + 1/7) T = 9$
 $T = 33 \text{ hrs. } 36 \text{ mins.}$
4. a. $60 \text{ km/hr} - 23 \text{ km/hr} = 37 \text{ km/hr} = \text{relative velocity}$
 $\text{Velocity} = \text{dist./time}$
 $37 = 100 / \text{time}$
 $\text{Time} = 2.702... \text{ hrs} = 2 \text{ hrs. } 42 \text{ min. } 9.7 \text{ sec}$
5. b. $60 + 23 = 83 \text{ km/hr} = \text{relative velocity}$
 $83 = 100 / \text{time}$
 $\text{Time} = 1.2048... \text{ hrs} = 1 \text{ hr. } 12 \text{ min. } 17.3 \text{ sec}$
6. e. $Y = 60 \text{ km/hr} * \text{time}$
 $X = 23 \text{ km/hr} * \text{time}$
 $Y^2 + X^2 = 100^2$
 $(60t)^2 + (23t)^2 = 100^2$
 $529t^2 + 3600t^2 = 100^2$
 $4129t^2 = 100^2$
 $t = 100/\sqrt{4129} \text{ hrs}$
7. d. $V - 15 = \text{velocity relative to the observers}$
 $(V-15)(60) = 20$
 $V = 15 \frac{1}{3} \text{ ft/s}$
8. c. $1 \text{ ft/s} = 10 \text{ ft} / \text{time1}$
 $\text{time1} = 10 \text{ s}$
 $\text{left for the rest of the distance} = 50 \text{ sec}$
 $V + 15 = \text{velocity relative to observers}$
 $(V+15)(50) = 10$
 $V = -14.8 \text{ ft/s}$
 Ans: they don't need to row/go faster; flow of water will help them beat the time/mark
9. c. Discharge. $Q = \text{volume/time}$
 $Q = \pi r^2 h/t$
 $= \pi(1)^2 3 / (5*60)$
 $= 0.01\pi \text{ ft}^3/\text{s}$
10. $Q = VA$
 $V = Q/A$
 $= (0.01\pi) / (1/4 * \pi * 0.25^2)$
 $= 0.64\pi^2 \text{ ft/s}$
11. d. $[1/12(8) + 1/9(6) + 1/11(5)] * \text{days} = 30$
 $\text{days} = 16.78 \text{ days} = 17 \text{ days consumed}$

12. b. $20 \times 19 \times 18 \times 17 = 116280$
Ans: 1 is to 116280 chances
13. c. Dist. Traveled by 1st plane, $A = 100 \times 6 = 600$ km north from origin
Dist. Traveled by 2nd plane, $B = 120 \times 6 = 720$ km 60° east of south
Cosine law:
(dist from each other) $^2 = A^2 + B^2 - 2AB \cos 60^\circ$
 $= 600^2 + 720^2 - 2 \times 600 \times 720 \times 1/2$
 $= 446400$
dist from each other $= \sqrt{446400}$ km $= 668.13$ km
14. e. $A = 100 \times 6 = 600$ km south from last position = origin
 $B = 60 \times 6 = 360$ km north from last position = directly east of origin
dist. from each other $= 720 \text{ km} \sin 60^\circ = 720 \times (\sqrt{3}/2) = 360 \sqrt{3} \text{ km} = 624 \text{ km}$

15. e. Right triangles:
Radius, $A = 3$ m
Distance, $C = \text{hypotenuse} = 5$ m
Other side, $B = 4$ m
 $\cos \theta = B/C = 4/5$
 $X = A \cos \theta$
 $= 3(4/5) = 12/5$
 $= 2.4 \text{ m} = \text{radius of what the cameraman sees}$
Diameter of what the cameraman sees $= 4.8 \text{ m}$



16. b. $X = 0.20 \text{ m}$
 $A = 3 \text{ m}$
 $\cos \theta = 0.20/3$
 $a^2 = X^2 + A^2$
 $= 0.20^2 + 3^2 = 9.04$
 $a = \sqrt{9.04}$
by similar triangles:
 $a/A = A/C$
 $\sqrt{9.04}/3 = 3/C$
 $C, \text{ distance of camera man} = 9/\sqrt{9.04} \text{ m} = 3 \text{ m}$

17. c. let $x = \text{number of Php0.25 coins}$
 $y = \text{number of Php1.00 coins}$
 $x + y = 128 \quad \text{---(1)}$
 $0.25x + y = 86 \quad \text{---(2)}$
 $(1) - (2) \quad 0.75x = 42 \quad \text{---(9/7)x = 42}$
 $x = 56$
 $y = 128 - 56 = 72$

18. a. let $x = \text{edad ni Ana ngayon}$
 $y = \text{edad ng anak ni Ana ngayon}$
 $x - 5 = 7(y - 5) \quad \text{---} x - 7y = -30 \quad \text{---(1)}$
 $x + 10 = 2(y + 10) \quad \text{---} x - 2y = 10 \quad \text{---(2)}$
 $(1) - (2) \quad -5y = -40$
 $y = 8$
 $x = 2y + 10 = 26$

19. a. let $c = \text{hrs Che can do the work alone}$
 $m = \text{hrs May can do the work alone}$
 $6/c + 6/m = 1 \quad \text{---(1)}$
 $3/m + 12/c = 1 \quad \text{---} 6/m + 24/c = 2 \quad \text{---(2)}$
 $(1) - (2) \quad c = 18, m = 9$

20. b. let $x = \text{halaga na in-invest sa 9\%}$
 $y = \text{halaga na in-invest sa 7\%}$
 $x + y = 1\,000\,000 \quad \text{---(1)}$
 $0.09x + 0.07y = 86\,000 \quad \text{---} (9/7)x + y = 86\,000 / 0.07 \quad \text{---(2)}$
 $(1) - (2) \quad x = 800\,000$

ENGLISH**Grammar**

1. e
2. b
3. e
4. c
5. b
6. c
7. e
8. b
9. c
10. b, illustrations,
11. e
12. b
13. d
14. c
15. a, ignored
16. c, has
17. a, finally,
18. b
19. e
20. d
21. b, 20-minute
22. d, two-storey
23. b, moved
24. e
25. a, were
26. a, came
27. c, to rest
28. a, toward
29. e
30. c, learning,
31. b, half-frozen
32. c, me
33. b, they're
34. b, lies

Context**Clues**

1. a
2. b
3. c
4. d
5. b
6. c
7. b
8. a
9. c
10. b
11. a
12. b
13. c
14. b
15. c
16. a
17. c
18. d
19. d
20. c
21. c
22. b
23. b
24. d
25. c
26. b
27. b
28. d
29. a
30. d
31. a
32. d
33. d
34. c
35. a

Reading

1. b
2. c
3. c
4. b
5. a
6. b
7. b
8. a
9. b
10. c
11. c
12. d
13. d
14. d
15. c
16. b
17. a
18. d
19. c
20. b

Analogy

1. c
2. d
3. b
4. e
5. e
6. b
7. d
8. e
9. b
10. c
11. a
12. d
13. a
14. a